

Resistance Exercise Sessions Comprising Multijoint vs. Single-Joint Exercises Result in Similar Metabolic and Hormonal Responses, But Distinct Levels of Muscle Damage in Trained Men

Paulo Henrique Barbosa,¹ Julio Benvenuti Bueno de Camargo,^{1,2} José Jonas de Oliveira,³ Carolina Gabriela Reis Barbosa,¹ Alessandro Santos da Silva,⁴ Julio Wilson Dos-Santos,⁵ Rozângela Verlengia,¹ Júlia Barreira,⁶ Tiago Volpi Braz,¹ and Charles Ricardo Lopes¹

¹Human Performance Research Laboratory, Methodist University of Piracicaba, Piracicaba, Brazil; ²Laboratory of Neuromuscular Adaptations to Resistance Training (MUSCULAB), Federal University of São Carlos, Brazil; ³Centro Universitário de Itajubá, Itajubá, Brazil; ⁴Postgraduate Program in Movement Sciences, São Paulo State University (UNESP), São Paulo, Brazil; ⁵Department of Physical Education, School of Sciences, São Paulo State University (UNESP), Bauru, Brazil; and ⁶University of Campinas, Campinas, Brazil

Abstract

Barbosa, PH, Bueno de Camargo, JB, Jonas de Oliveira, J, Reis Barbosa, CG, Santos da Silva, A, Dos-Santos, JW, Verlengia, R, Barreira, J, Braz, TV, and Lopes, CR. Resistance exercise sessions comprising multijoint vs. single-joint exercises result in similar metabolic and hormonal responses, but distinct levels of muscle damage in trained men. *J Strength Cond Res* 38(5): 842–847, 2024—Resistance-type exercise (RE) elicits distinct acute metabolic and hormonal responses, which can be modulated by the manipulation of training variables. The purpose of this study was to compare the metabolic (blood lactate and estimated lactic anaerobic system energy expenditure) and hormonal (growth hormone [GH]) responses to RE sessions composed exclusively of multijoint (MULTI) or single-joint (SINGLE) exercises. Assessments of creatine kinase (CK) levels were also performed. In a crossover design, 10 recreationally resistance-trained men (age: 26.9 ± 3.0 years, total body mass: 83.2 ± 13.8 kg; height: 176 ± 7.0 cm; training experience: 5.5 ± 2.4 years) were randomly submitted to both protocols. Blood collections were made pre, 3 minutes after, and 36 hours after each experimental session. No significant difference between MULTI vs. SINGLE was observed for the rises in blood lactate ($p = 0.057$) and GH ($p = 0.285$) levels. For CK, a significant difference between the protocols was noted, in which MULTI resulted in significant rises after 3 minutes ($p = 0.017$) and 36 hours ($p = 0.043$) compared with SINGLE. In conclusion, the findings of this study suggest that resistance-trained individuals display similar metabolic and hormonal responses when performing MULTI and SINGLE exercise protocols. Also, RE sessions comprising MULTI exercises induce a higher magnitude of muscle damage, which may require a longer recovery period compared with SINGLE.

Key Words: strength training, growth hormone, blood lactate, energy expenditure, creatine kinase

Introduction

Resistance-type exercise (RE) induces several acute metabolic and hormonal responses, which may present an important role in the subsequent skeletal tissue remodeling (19). Different RE programs regarding intensity (23,24), volume (14), and muscle actions (9) have been shown to result in distinct acute hormonal and metabolic responses. According to the practitioner's goal or demand, the type of exercise performed is another training variable that is usually manipulated by strength and conditioning coaches. Generally, RE can be labeled according to the number of joints involved in a specific movement as multijoint (MULTI) or single joint (SINGLE) (12).

Although the type of exercise performed may eventually affect the acute responses to a given training session, this topic is still underinvestigated, especially in resistance-trained individuals. For blood lactate levels, recent data from our research group

reported that trained men experienced significantly higher increases when performing a MULTI exercise (barbell bench press) compared with a SINGLE exercise (dumbbell fly) (3), which may be mainly explained by the higher volume load (VL) observed during the first.

It has been extensively reported that distinct RE protocols may influence the magnitude of acute hormonal responses (e.g., growth hormone [GH]) (14,20,28). Although RE protocols with higher total work and shorter rest intervals have been shown to induce the larger rises in GH, eventual influences of the type of exercise performed remains to be properly investigated.

In addition to metabolic disturbances, RE can also result in localized damage to the muscle tissue, which is usually accompanied by the release of enzymes such as creatine kinase (CK) into the blood, reductions in neuromuscular performance-related parameters, and also acute morphological responses (e.g., muscle swelling, limb circumference) (18). The effects of different types of RE (MULTI vs. SINGLE) on indirect markers of muscle damage are not completely elucidated, with conflicting results being reported. Soares et al. (27), for example, observed that performing an exercise session exclusively composed of a SINGLE

Address correspondence to Júlio Benvenuti Bueno de Camargo, julio.bbc@gmail.com.

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exercise (biceps curl) elicited higher delayed-onset muscle soreness (DOMS) and isometric strength reductions compared with MULTI (lat pull-down). Conversely, in a recent study from our research group (7), performing the back squat exercise (MULTI) resulted in a higher-magnitude elevation in indirect markers of muscle damage (limb circumference, muscle swelling, DOMS) compared with the knee extension exercise (SINGLE). These distinct results may be mainly explained by the fact that only protocols for the upper and lower limbs were adopted for Gentil et al. (12) and De Camargo et al. (7), respectively. In this sense, it is not completely clear whether the responses of the indirect markers of muscle damage assessed in each of these studies were resulted by the muscle groups stimulated within the session (upper-limb vs. lower-limb muscles) or the type of exercise (MULTI vs. SINGLE) itself. Therefore, experimental protocols, including both upper-limb and lower-limb exercises, within the same training session may help to provide further insight into this topic.

It is interesting to note that based on these investigations, although both metabolic and indirect markers of muscle damage outcomes may somewhat differ between MULTI and SINGLE exercises, they were limited to assessing these changes in exercise bouts that involved only 1 exercise, which impairs possible inferences toward training sessions performed by trained individuals in their usual training routines. Therefore, this study aimed to assess the acute metabolic (blood lactate, GH levels, and estimated anaerobic energy cost) and indirect muscle damage (CK levels) responses to performing MULTI vs. SINGLE exercise sessions. It was initially hypothesized that because of the higher amount of muscle mass involved and the magnitude of VL, the MULTI protocol would induce higher responses in all the dependent variables assessed compared with the SINGLE.

Methods

Experimental Approach to the Problem

This study followed a randomized crossover design. Subjects visited the laboratory on 5 separate occasions (Figure 1). The first visit consisted of collecting the Physical Activity Readiness Questionnaire (PAR-Q), anthropometric data, 1-repetition maximum (1RM) test (bench press and back squat exercises) followed by an 8 RM test (interspaced by 30 minutes) for all the exercises adopted during the experimental protocols. In the second visit (after 72 hours), MULTI or SINGLE exercise protocols were randomly performed, and 1 week later, each subject was submitted to the opposite protocol. Blood collections were made before, 3 minutes after, and 36 hours after each exercise session. The MULTI protocol consisted of the bench press, lat pull-down, and back squat exercises, and the SINGLE protocol consisted of the dumbbell fly, straight arm cable pull-down, and leg extension exercises. A 1-minute and 3-minute passive rest interval was afforded between sets and exercises, respectively. All subjects were instructed to keep their usual nutrition habits and to refrain from any exercise for 1 week before performing the experimental protocols.

Subjects

Ten recreationally resistance-trained men (age: 26.9 ± 3.0 years [range 19–34 years]; total body mass: 83.2 ± 13.8 kg; height: 176 ± 7.0 cm; training experience: 5.5 ± 2.4 years [range 2–5 years]; training frequency: 5.3 ± 0.7 sessions per week; relative back squat 1RM: 1.4 ± 0.3) participated in the study. All subjects

performed RE for a minimum of 3 days per week for at least 1 year and the exercises adopted in the experimental sessions (minimum frequency of once a week). In addition, subjects did not present any existing musculoskeletal disorders and negatively answered all questions on the PAR-Q. Subjects also displayed a minimum relative 1RM value for the squat and bench press exercises of at least 1.25 and $1.0 \times$ total body mass, respectively. This study was approved by the Methodist Piracicaba University's research ethics committee (protocol 13/19); all subjects read and signed an informed consent document.

Procedures

Experimental Acute Protocols. In random order (block randomization by a researcher not involved in the data collection process), subjects were submitted to both MULTI and SINGLE exercise protocols. During both sessions, they were instructed to perform a maximal number of repetitions until the point of concentric muscular failure with a load corresponding to 8 RM for 5 sets. Subjects reported a rating of perceived exertion (RPE) based on the repetitions in reserve (RIR) scale of 9.5–10 for all sets and exercises across both experimental sessions (17). At baseline (1 week before the experimental sessions), an 8RM test for all the exercises adopted was performed within 3 attempts, with a rest interval of 5 minutes between the attempts and a 10-minute recovery period between exercises. The testing protocol followed previous recommendations by the National Strength and Conditioning Association (NSCA) (26). The heaviest load achieved on 3 attempts was used to set the initial training load. Before the first set of each experimental condition, the subjects performed a specific warm-up, consisting of 8 repetitions at 50% of the 8RM. The loads were continually adjusted to keep the exercises in 8RM. A standard cadence of 4 seconds per repetition was adopted using a metronome (Metronome Beats, Stonekick, London, United Kingdom). A passive rest period of 60 second was adopted between the sets. Both protocols were accompanied by the same researchers. All volunteers received verbal encouragement during the exercises.

Maximal Dynamic Strength (1RM). Maximum dynamic strength was assessed through 1RM testing using the back squat ($1RM_{\text{SQUAT}}$) and bench press ($1RM_{\text{BENCH PRESS}}$) exercises. This test was performed to characterize the relative maximum dynamic strength of the sample, ensuring that all subjects displayed a minimum $1RM_{\text{SQUAT}}$ and $1RM_{\text{BENCH}}$ of 1.25 and $1.0 \times$ body mass, respectively (2). The testing protocol followed previous recommendations by the NSCA (26). During the first set, subjects performed 5 repetitions at 50% of the estimated 1RM followed by 1 set of 3 repetitions at a load corresponding to 60–80% of the estimated 1RM with a 1-minute rest interval between the sets. After the warm-up sets, subjects had 3 attempts to find their 1RM load with 3-minute intervals between the trials. The 1RM was deemed as the maximum weight that could be lifted no more than once with the proper technique. All testing sessions were supervised by the same researchers.

Blood Lactate. Blood lactate concentration was assessed to estimate the anaerobic energy expenditure of each training session (post 3 minutes). Blood samples ($25 \mu\text{l}$) were collected from the fingertips into heparinized capillary tubes and transferred to microtubes containing $50 \mu\text{l}$ of 1% sodium fluoride. The lactate concentration was analyzed through an electroenzymatic method with a lactate analyzer (YSI 2300 Stat Analyzer, Yellow Springs

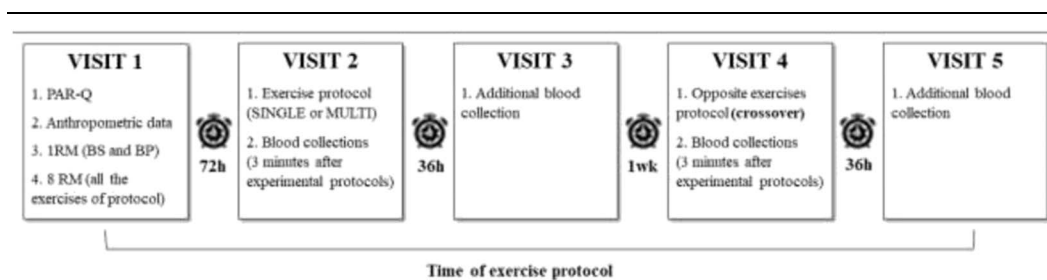


Figure 1. Experimental design of the study. BS = back squat; BP = bench press; h = hours; wk = week.

Instruments, 2 Yellow Springs, OH). The values are expressed in millimoles $\times$ liter⁻¹ (mmol $\cdot$ L⁻¹).

Growth Hormone and Creatine Kinase Levels. Blood samples (~5 ml) were obtained by a standard venipuncture in 2 vacutainer tubes (Becton Dickinson, Juiz de Fora, Brazil), one containing heparin for plasma separation and the other for serum separation. After allowing to clot at room temperature for 30 minutes, both tubes were centrifuged at 2,000 rpm for 20 minutes at 4° C, and the plasma and serum samples were stored at -70° C for later analyses. Serum GH was assessed by radioimmunoassay and values are expressed in nanograms $\times$ liter⁻¹ (ng $\cdot$ L⁻¹). Serum CK activity was measured using an automated equipment Konelab 60i (Wiener Lab, Rosario, Argentina) at 37° C using a test kit (Wiener Lab), and values are expressed as international units $\times$ liter⁻¹ (IU $\cdot$ L⁻¹).

Estimated Anaerobic Energy Expenditure. According to the methodology previously described by Bertuzzi et al. (1), the GEDAE-LaB software (<http://www.gedaelab.org/>) was adopted to estimate the anaerobic lactic energy system contribution, in which the lactate accumulation (postexercise lactate minus resting lactate) was multiplied by the oxygen equivalent (3 ml $\cdot$ O₂ $\cdot$ kg⁻¹) (8) and by the subject's body mass. The data are expressed in kilocalories (Kcal).

Volume Load. The VL (sets $\times$ repetitions $\times$ load [kg]) was calculated from training logs filled out by research assistants for each one of the experimental sessions (MULTI and SINGLE) (11). The values are expressed in kilograms (kg).

Statistical Analysis

The normality and homogeneity of the variances were verified using the Shapiro-Wilk and Levene's tests, respectively. Before analysis, all data were log transformed for analysis to reduce bias arising from non-uniformity error (heteroscedasticity). The mean, SD, and 95% confidence intervals (CI) were used after data normality was assumed. A 2 $\times$ 2 repeated-measures analysis of variance (ANOVA; interaction conditions [MULTI and SINGLE] $\times$ time [pre vs. post]) was used to compare the blood lactate and GH analysis. A 2 $\times$ 3 repeated-measures ANOVA (interaction conditions [MULTI and SINGLE] $\times$ time [pre, post, and post 36 hours]) was used to compare CK values. Post hoc comparisons were performed with the Bonferroni's correction. Assumptions of sphericity were evaluated using Mauchly's test. Where sphericity was violated ($p < 0.05$), the Greenhouse-Geisser correction factor was applied. In addition, an unpaired t test was used to compare VL, absolute

differences in blood lactate, GH, and CK values and also anaerobic energy contribution between the protocols. The effect size (ES) in ANOVA was evaluated using a partial eta-squared (η_p^2) value of <0.06 (small), 0.06–0.14 (medium), and >0.14 (large) (6). The ES in absolute difference (pre vs. post) was based on Cohen's d units by means (d value) (6). The ES result was qualitatively interpreted using the following thresholds: <0.2 , trivial; 0.2–0.6, small; 0.6–1.2, moderate; 1.2–2.0, large; 2.0–4.0, very large; and >4.0 , extremely large. All analyses were conducted in SPSS-22.0 software (IBM Corp., Armonk, NY). The adopted significance was $p < 0.05$.

Results

For blood lactate, a significant main effect of time ($F_{1,36} = 156.7$; $\eta_p^2 = 0.81$), but not group $\times$ time interaction ($F_{1,36} = 3.939$; $\eta_p^2 = 0.09$), was observed (Table 1). The ES of the absolute difference (pre-post) was extremely large for SINGLE ($d = 4.79$) and very large for MULTI ($d = 3.75$). The ES between the protocols was moderate ($d = 0.91$).

For GH, a significant main effect of time ($F_{1,36} = 30.23$; $\eta_p^2 = 0.46$), but not group $\times$ time interaction ($F_{1,36} = 10.82$; $\eta_p^2 = 0.03$), was observed (Table 1). The ES of the absolute difference (pre-post) was large for both SINGLE ($d = 1.55$) and MULTI ($d = 1.96$). The ES between the protocols was small ($d = 0.49$).

Figure 2 displays CK values pre, post 3 minutes, and post 36 hours for each experimental protocol. A significant main effect of time for MULTI ($F_{2,48} = 13.18$; $\eta_p^2 = 0.35$), but not group $\times$ time interaction ($F_{2,48} = 23.57$; $\eta_p^2 = 0.05$), was observed. The ES of the absolute difference (pre to 3 minutes) was small for SINGLE ($d = 0.45$) and large for MULTI ($d = 1.49$), and the ES of the absolute difference (pre to 36 hours) was moderate for SINGLE ($d = 0.86$) and large for MULTI ($d = 1.80$). The ES between the protocols was large for pre to 3 minutes ($d = 1.16$) and moderate for pre to 36 hours ($d = 0.97$).

No significant difference was observed between the protocols for the estimated anaerobic energy cost ($p = 0.075$; mean difference [MD] = 3.10; 95% CI = -0.34 to 6.54 kcal; Figure 3). Figure 4 displays the comparison of the absolute difference (post - pre) between the protocols for the dependent variables assessed. No significant difference between MULTI vs. SINGLE was observed for the rises in blood lactate ($p = 0.057$; MD = 2.59, 95% CI = -0.09 to 5.27 mmol $\cdot$ L⁻¹) and GH levels ($p = 0.285$, MD = 0.93, 95% CI = -0.82 to 2.73 ng $\cdot$ L⁻¹). For CK, a significant difference between the protocols was noted, in which MULTI resulted in significantly higher rises after 3 minutes ($p = 0.017$; MD = 94.4, 95% CI = 18.3–170.5 IU $\cdot$ L⁻¹) and 36 hours ($p = 0.043$; MD = 146.7, 95% CI = 5.18–288.2 IU $\cdot$ L⁻¹) compared with SINGLE.

Table 1**Pre and post values for blood lactate and growth hormone (mean $\pm$ SD).***

Variables	Pre	Post	MD [95% CI]	Time <i>p</i>	Time $\times$ group <i>p</i>
Lactate (mmol·L ⁻¹)					
SINGLE	1.19 $\pm$ 0.17	8.11 $\pm$ 2.03	6.92 [4.4–9.3]	0.001	0.054
MULTI	1.36 $\pm$ 0.40	10.87 $\pm$ 3.56	9.51 [7.0–11.9]	0.001	
GH (ng·L ⁻¹)					
SINGLE	0.04 $\pm$ 0.04	1.97 $\pm$ 1.75	1.93 [0.2–3.6]	0.016	0.305
MULTI	0.04 $\pm$ 0.03	2.90 $\pm$ 2.06	2.86 [1.1–4.5]	0.001	

*SINGLE = single-joint protocol; MULTI = multijoint protocol; GH = growth hormone; MD (95% CI) = mean difference (95% confidence interval); *p* = *p* value.

Discussion

The aim of this study was to compare the acute metabolic and hormonal responses between MULTI vs. SINGLE training sessions, which comprised exercises for both upper and lower limbs. In addition, the levels of an indirect marker of muscle damage (CK) were also assessed. Our initial hypothesis was partially confirmed because both blood lactate and GH rises did not differ between the exercise protocols, but the MULTI session did result in significant CK increases post 36 hours.

Resistance exercise is an effective physiological stimulus for increasing both GH and (29) and blood lactate levels (5). Although both exercise bouts induced significant rises in plasma GH and lactate levels, no difference was noted between the protocols. Our initial hypothesis was that because of the larger amount of muscle mass involved and total work performed, MULTI would induce a higher rise in postexercise serum GH and lactate. The acute responses of both variables are strongly influenced by RE protocols with higher volumes (14,15,16,19,21). Hansen et al. (16), for example, reported that the higher VL observed when performing MULTI (leg press) in addition to a SINGLE exercise (biceps curl) induced higher rises in GH compared with an exclusively SINGLE session. This effect on acute GH responses elicited by exercise sessions with higher VL was also reported by other investigations (14,15). Notwithstanding, similar to another investigation (20), the larger VL observed by this study for the MULTI compared with the SINGLE protocol (7,302 $\pm$ 1,284 kg vs. 5,590 $\pm$ 1,054 kg, respectively) was not sufficient to induce significantly larger rises in serum GH and lactate levels. Therefore, although somewhat speculative, there may exist a threshold of total work performed that is required to result in significant differences in both hormonal and metabolic responses throughout the early recovery period, especially for previously trained subjects. Additional studies are

warranted to provide further insight on to which extent the total work performed is related to these responses following distinct RE protocols. Moreover, because this is the first controlled study that evaluated the effect of MULTI vs. SINGLE exercise protocols on GH and lactate responses, we cannot directly compare our findings with other studies.

Although the exact role of exercise-induced metabolites and hormonal rises in mediating muscular responses to RE programs is far from being elucidated, GH release seems to be somehow associated with blood lactate levels. This relationship arises from studies that reported a blunted exercise-induced GH response in healthy subjects following the administration of alkaline compounds (sodium bicarbonate) (10) and also in patients with McArdle's disease, which present an impaired capacity for elevating blood lactate levels within exercise testing of progressive load (13). Therefore, the lacking of difference in GH response between MULTI and SINGLE protocols in this study may also be justified by the similar increases in blood lactate levels induced by both experimental conditions.

In addition to blood lactate levels, this study also assessed anaerobic lactic pathway–estimated energy expenditure as an indirect measure of glycolytic activity, with no significant difference emerging between the experimental protocols. Eventual comparisons with other studies cannot be properly done because this is the first investigation that compared distinct RE sessions regarding the type of exercises performed (MULTI vs. SINGLE) and their acute effects on the lactic metabolism contribution. Our initial hypothesis was that because of a strong and positive association between total work performed and anaerobic energy cost within a RE session (25), the larger VL performed during MULTI would result in an increased energetic contribution of the anaerobic system compared with the SINGLE session. Brunelli et al. (4), for example, reported a higher contribution of the anaerobic energetic pathway following an RE

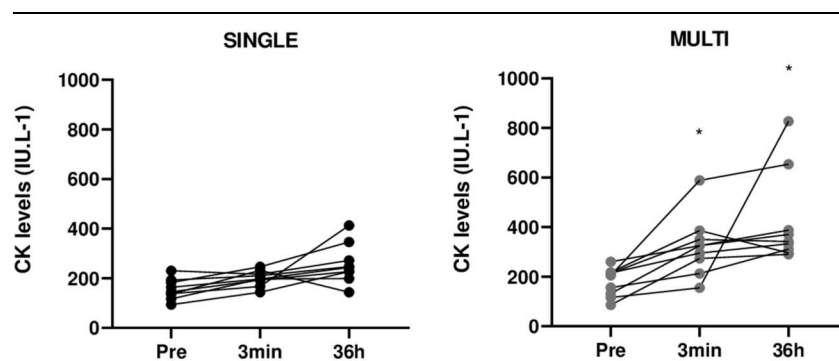


Figure 2. Creatine kinase (CK) pre, 3 minutes after, and 36 hours after each training session for SINGLE and MULTI interventions. SINGLE = single-joint protocol; MULTI = multijoint protocol; *Significantly different compared with the premoment (*p* < 0.001).

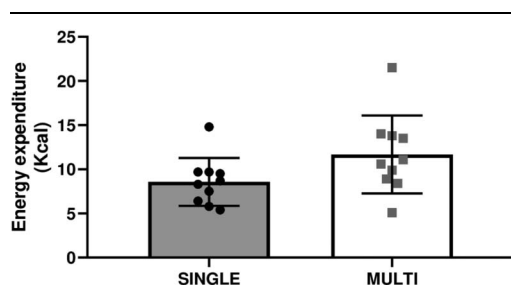


Figure 3. Estimated anaerobic lactic energy system contribution; SINGLE = single-joint protocol; MULTI = multijoint protocol.

protocol with larger VL compared with a smaller one. However, it is of great relevance to highlight that, different from the current investigation, Brunelli et al. (4) included only subjects with no previous training experience. In addition, the same exercise (leg extension) performed with different relative intensity and also distinct VL values was performed. Therefore, based on our findings, although a trend was observed favoring the MULTI protocol ($p = 0.075$), it can be suggested that the contribution of the anaerobic lactic system during an RE session is not dependent on the exercise performed as long as a high-intensity protocol is performed to the point of momentary concentric muscular failure. The authors of this study strongly encourage that additional investigations be performed to further elucidate our findings and also help strength and conditioning coaches to optimally manipulate the type of exercise performed within RE sessions according to specific goals or demands of each practitioner.

For CK levels, significant increases from pre to post 36 hours were noted only for the MULTI protocol. Figure 2 shows a greater increase in CK levels for 2 individuals in the MULTI protocol, reinforcing the individual variability of this marker after the

resistance training. After performing the statistical analysis without these 2 cases, we found the same results corroborating that the findings are not only dependent on these 2 individuals but rather represent the group. Although not completely understood, this may be mainly attributed to the fact that the accumulated volume of work seems to be the primary determinant of exercise-induced CK rises in trained subjects (22). This result confirms previous findings from our laboratory (7), in which performing exclusively the back squat exercise elicited higher post-36 hour rises in indirect markers of muscle damage (leg circumference, muscle swelling, muscle soreness) compared with performing the knee extension exercise. Different from this study, Soares et al. (27) reported that a SINGLE protocol resulted in higher-magnitude rises in muscle damage markers compared with a MULTI one. Although the subjects presented similar training backgrounds, some methodological differences must be acknowledged to compare the distinct results between Soares et al. (27) and the current study. First, neuromuscular performance (elbow flexor peak torque) and muscle soreness-related outcomes were assessed as indirect markers of muscle damage by Soares et al. (27). In this sense, it is important to highlight that post-exercise rises in CK levels are not necessarily associated with other markers of muscle damage, such as muscle strength and soreness (18). Second, 36 hours postexercise assessments were not performed by Soares et al. (27). Moreover, although this investigation assessed the responses to an exercise session composed of 3 exercises (lower and upper limbs), only 1 exercise was performed in each protocol by Soares et al. (27) (unilateral seated row and biceps curl for MULTI and SINGLE, respectively). Therefore, from an entire training session standpoint, further investigations must be addressed to better understand the responses of indirect markers of muscle damage to MULTI vs. SINGLE exercise protocols.

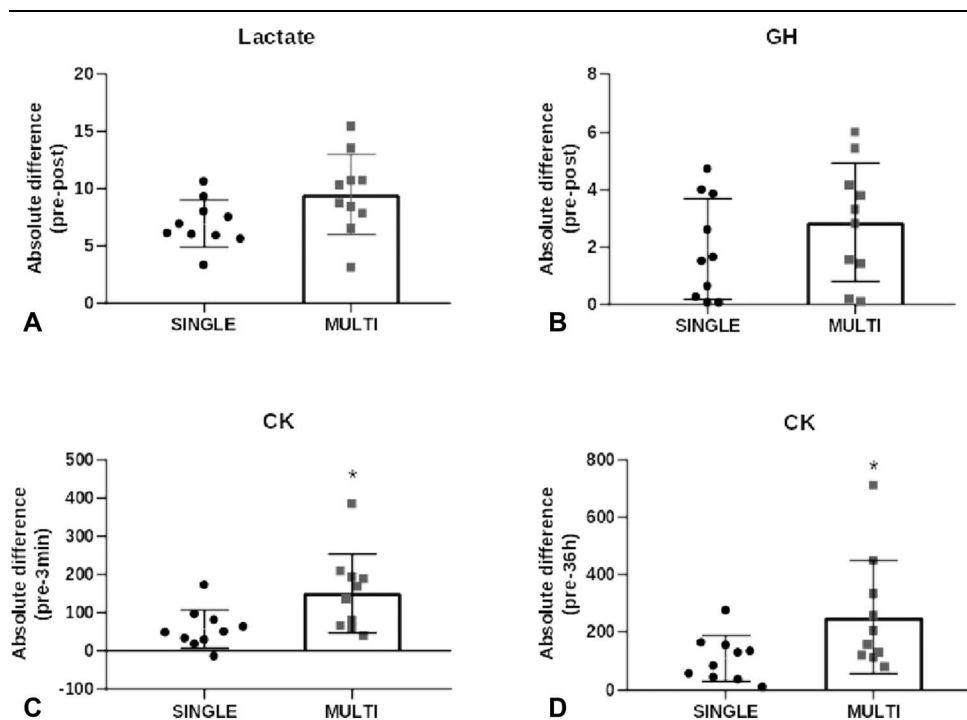


Figure 4. Absolute different (pre-post) for blood lactate (A), growth hormone (B), pre to 3 minutes (C), and pre to 36 hours (D) for creatine kinase. SINGLE = single-joint protocol; MULTI = multijoint protocol; GH = growth hormone; CK = creatine kinase. *Significantly different from SINGLE.

The main limitation of this study is that no further analysis was performed during the later phases of the recovery period (e.g., 48, 72, 96 hours postexercise sessions). Therefore, it is unclear whether similar metabolic and muscle damage patterns would be observed throughout additional assessments. It is also important to highlight that the results reported in this study refer only to recreationally trained men. In this sense, future studies with other populations (e.g., untrained, women, older subjects) must be encouraged to provide a better understanding of this topic. Finally, inferences regarding muscle damage must be done with caution because only CK levels were assessed. The validity of the analysis of CK for reflecting exercise-induced skeletal muscle damage is still controversial within the literature because factors such as training level and gender can influence CK levels to a greater extent than differences in exercise volume completed (18). Therefore, future studies investigating other indirect markers of muscle tissue damage such as neuromuscular performance, morphological parameters (e.g., muscle swelling, limb circumference), and muscle soreness-related parameters must be conducted to provide further insight into this topic.

In conclusion, the results of this study suggest that the magnitude of muscle tissue damage seems to be influenced by the type of exercise performed within an exercise bout by recreationally resistance-trained individuals. In addition, MULTI and SINGLE protocols result in similar acute metabolic and hormonal responses.

Practical Applications

The information provided by the current investigation may be of great relevance for strength and conditioning coaches when prescribing and implementing strength training programs. From a practical standpoint, both protocols are feasible to be implemented within high-intensity RE programs that aim to elevate both metabolic and hormonal stress. Also, based on our findings, it can be suggested that RE sessions comprising MULTI exercises induce a higher magnitude of skeletal muscle tissue damage, which may require a longer recovery period compared with SINGLE. Future studies must be conducted to investigate whether this increased stress observed in the MULTI protocol would result in augmented chronic adaptations. Finally, the results of this study also suggest that exercise sessions comprising exclusively SINGLE exercises may be adopted in specific contexts that require a reduced stressful stimulus, such as sessions performed within precompetition phases.

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